Baibhab Karmakar

Master's Student in Earth Sciences, IISER Kolkata, Kolkata, India
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RESEARCH INTERESTS

Over the past two years, I have worked on earthquake source characterization, slip inversion, surface-wave amplification, and machine learning-based earthquake relocation, using observational and computational approaches to analyse seismic data. I aim to deepen my expertise through PhD research in understanding earthquake processes, seismic wave propagation, and Earth structure, with a particular interest in computational geophysics and data-driven methods in seismology.

EDUCATION

Indian Institute of Science Education and Research (IISER) Kolkata, Kolkata, India — 2026 – Present
Master's Student in Earth Sciences
Thesis Title: Earthquake Relocation Using Machine Learning

Indian Institute of Science Education and Research (IISER) Kolkata, Kolkata, India — 2022 – 2026
BS-MS, Earth Sciences Major and Computer Science Minor — CGPA: 8.09/10.00

Higher Secondary Examination — 2022
Score: 95.8%

Madhyamik Examination — 2020
Score: 96.29%

RESEARCH EXPERIENCE

Computational Seismology Laboratory, IISER Kolkata — Kolkata, India
Master's Researcher, under Prof. Supriyo Mitra — 2026 — Present

- Applying QSeek, a machine-learning-based framework for earthquake detection and localization, to continuous seismic data from the Kashmir Himalaya.
- Integrating neural-network-based phase pickers, including PhaseNet and EQTransformer from SeisBench, with stacking-migration and adaptive octree localization to develop a more comprehensive earthquake catalog for the region.
- Evaluating the QSeek-derived catalog against the existing regional catalog to assess improvements in detection completeness and location accuracy, and to better understand the spatial and temporal patterns of seismicity in the Kashmir Himalaya.

Institut de Physique du Globe de Paris (IPGP) — Paris, France
Summer Research Intern, under Dr. Martin Vallée — June 2026 — July 2026

- Performed joint kinematic slip inversions for the 2015 and 2025 Charlie-Gibbs Transform Fault earthquakes using teleseismic P/SH waveforms and moment-constrained Empirical Green's Function relative source time functions (RSTFs), resolving unilateral westward rupture propagation in both events.
- Revealed contrasting rupture patterns on the Charlie-Gibbs fault, with the 2015 earthquake dominated by a localized high-slip asperity, while the 2025 event exhibited more distributed slip and limited re-slip at the 2015 peak-slip region.
- Improved the seismic-data fit by incorporating spatially variable fault dip as an inversion parameter, while noting that the resulting dip distribution lacks independent observational constraints.

IISER Pune — Pune, India
Research Intern — 2025

- Investigated surface-wave amplification across the Shillong Plateau–Bengal Basin boundary using 1D shear-wave velocity profiles, focusing on the effects of low shear velocities and impedance contrasts at the basin edge.
- Modeled surface-wave propagation using SWRT (semi-analytical solutions across transversely aligned vertical discontinuities, incorporating higher-mode conversion) alongside SPECFEM2D full-waveform simulations; both methods predicted 2–5× higher amplitudes in the basin than in crystalline terrain over 0.01–0.15 Hz.
- Validated model predictions against a regional earthquake recorded at both sites, finding good agreement on the transverse component; presented as a poster at AGU Annual Meeting 2025.

Environmental Science Laboratory, IISER Kolkata — Kolkata, India
Research Assistant — 2024

- Assisted in environmental data collection and analysis on pesticide contamination in milk samples.

TEACHING EXPERIENCE

IISER Kolkata Kolkata, India
Teaching Assistant - Geophysics and Hydrology Autumn 2026 - Present

- Assisting in tutorials and practical sessions for undergraduate students.
- Supporting students in conceptual understanding and problem-solving in geophysics and hydrology.

IISER Kolkata Kolkata, India
Teaching Assistant - Introduction to Earth Science Autumn 2025

- Assisted in conducting tutorials and practical sessions for undergraduate students.
- Supported grading, doubt-clearing sessions, and conceptual understanding of core Earth Science topics.

PROJECTS

Receiver Function Tool IISER Kolkata, India
Self-directed Computational Project 2025

- Built an automated and interactive tool for crustal thickness estimation using receiver-function analysis.
- Developed workflows for seismic interpretation and theoretical travel-time modelling.

PUBLICATIONS

Journal paper
In Preparation

- Karmakar, B., Vallée, M.. Ten-year fault reactivation on the Charlie-Gibbs Transform Fault: joint kinematic inversion and slip asperity distribution analysis of the 2015 and 2025 earthquakes (working title, in preparation).

PRESENTATIONS

Poster Presentation - AGU Annual Meeting 2025
American Geophysical Union (AGU)

- Presented my work on seismic surface wave amplification in the Bengal Basin.
- Discussed observational and modelling results with an international research audience.

SELECTED COURSES

Master's Courses

- Master's Thesis Research in Computational Seismology

Additional Courses

- Data Science and Machine Learning
- Artificial Intelligence for Data Science

Bachelor's Courses

- Seismology
- Seismology Lab
- Geophysics
- Geodynamics
- Structural Geology
- Waves and Optics

AWARDS & FELLOWSHIPS

Innovation in Science Pursuit for Inspired Research (INSPIRE) India
Department of Science and Technology, Government of India 2022 – Present

JBNSTS Junior Fellow India
Jagadis Bose National Science Talent Search 2020 – 2022

SKILLS

- **Programming:** Python, C, MATLAB, Shell scripting
- **Python Libraries:** NumPy, Pandas, SciPy, Matplotlib, ObsPy, Seaborn, Scikit-learn, SeisBench, TensorFlow and etc.
- **Geophysical Software:** SAC, SPECFEM2D, Pyrocko, SWRT, ASPECT(basic)
- **Machine Learning:** Supervised & Unsupervised Learning (Regression, SVM, k-NN, Naive Bayes, Decision Trees, k-Means, PCA, DBSCAN), Ensemble Methods (Random Forest)
- **Deep Learning:** Neural Networks (Perceptron, MLP,), Optimization (Gradient Descent, Adam)
- **Other:** Data Structures and Algorithms, Git, LaTeX, Microsoft Office Suite and etc.

REFERENCES

Prof. Supriyo Mitra
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Scholar Profiles: Personal Page - Google Scholar

Dr. Martin Vallée
Researcher, Institut de Physique du Globe de Paris (IPGP), Paris, France
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Dr. Arjun Datta
Assistant Professor, Department of Earth and Climate Science, Indian Institute of Science Education and Research (IISER) Pune, Pune, India
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